

(12) **United States Patent**
Guo et al.

(10) **Patent No.:** **US 12,415,562 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **WHEELED CARRYING APPARATUS**

(56) **References Cited**

(71) Applicant: **Wonderland Switzerland AG**,
Steinhausen (CH)

U.S. PATENT DOCUMENTS

(72) Inventors: **Zheng-Wen Guo**, Guangdong (CN); **Er Xue Wang**, Guangdong (CN);
Shoufeng Hu, Guangdong (CN);
Mingxing Sun, Guangdong (CN);
Hongtao Yin, Guangdong (CN)

4,349,938 A * 9/1982 Fontana B60B 33/02
16/35 R
4,779,879 A * 10/1988 Kassai B62B 7/08
280/643

(Continued)

FOREIGN PATENT DOCUMENTS

(73) Assignee: **Wonderland Switzerland AG**,
Steinhausen (CH)

CA 2756004 A1 10/2010
CN 1031204 A 2/1989

(Continued)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

OTHER PUBLICATIONS

(21) Appl. No.: **18/762,108**

2nd Office Action for Corresponding Chinese Patent Application
No. 2020109515460 dated Sep. 4, 2024.

(22) Filed: **Jul. 2, 2024**

(Continued)

(65) **Prior Publication Data**

US 2024/0351626 A1 Oct. 24, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/945,323, filed on
Sep. 15, 2022, now Pat. No. 12,172,693, which is a
(Continued)

Primary Examiner — Jeffrey J Restifo

(74) *Attorney, Agent, or Firm* — Volpe Koenig

(30) **Foreign Application Priority Data**

Sep. 11, 2019 (CN) 201910858955.3
Apr. 30, 2020 (CN) 202010370191.6

(57) **ABSTRACT**

A wheeled carrying apparatus includes a standing frame, a handle frame, a wheel mount, a locking assembly and a release mechanism. The handle frame is rotatable between a first and a second position. The wheel mount carries a wheel and is rotatable relative to the standing frame for changing an orientation of the wheel axis. The locking assembly is configured to rotationally lock the wheel mount with respect to the standing frame, and configured to unlock the wheel mount so that the wheel mount is rotatable relative to the standing frame for changing an orientation of the wheel axis. The release mechanism includes a first portion provided on the standing frame and coupled to the locking mechanism, and a second portion carried with the handle frame, the second portion being movable along with the handle frame toward and away from the first portion during rotation of the handle frame.

(51) **Int. Cl.**
B62B 9/08 (2006.01)
B62B 9/12 (2006.01)

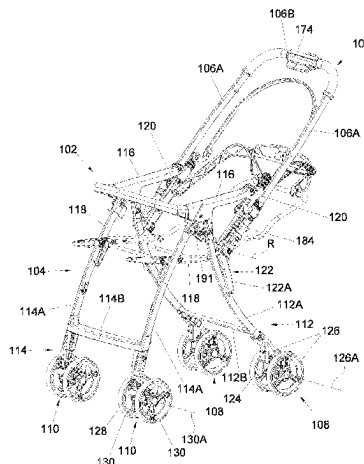
(52) **U.S. Cl.**
CPC . **B62B 9/08** (2013.01); **B62B 9/12** (2013.01)

(58) **Field of Classification Search**
CPC B62B 5/0438; B62B 7/042; B62B 9/08;
B62B 9/085; B62B 9/087; B62B 9/203;
B60B 33/026

See application file for complete search history.

14 Claims, 12 Drawing Sheets

100



Related U.S. Application Data

continuation of application No. 17/017,401, filed on
Sep. 10, 2020, now Pat. No. 11,479,285.

2022/0009538 A1* 1/2022 Guo B62B 9/087
2022/0266887 A1 8/2022 Guo
2023/0059506 A1* 2/2023 Wu B60B 33/045

FOREIGN PATENT DOCUMENTS

(56)	References Cited				
	U.S. PATENT DOCUMENTS				
	4,997,066	A *	3/1991	Bigo	B62B 9/082
					188/69
	5,125,676	A *	6/1992	Teng	B60B 33/00
					280/47.38
	5,351,364	A *	10/1994	Zun	B60B 33/025
					16/35 R
	6,212,733	B1 *	4/2001	Yeh	B60B 33/0002
					16/35 R
	6,315,309	B1 *	11/2001	Li	B62B 7/08
					280/47.38
	7,083,175	B1 *	8/2006	Liu	B60B 33/0039
					280/47.38
	7,210,690	B2 *	5/2007	Tan	B62B 9/18
					280/47.38
	7,273,225	B2	9/2007	Yeh	
	7,490,848	B2 *	2/2009	Wu	B62B 7/04
					280/47.38
	7,922,182	B2 *	4/2011	Van der Vegt	B62B 7/068
					280/47.39
	8,328,226	B2 *	12/2012	Xu-Hui	B62B 7/042
					280/658
	8,807,587	B2 *	8/2014	Funakura	B62B 7/04
					280/47.36
	8,991,854	B2	3/2015	Greger et al.	
	9,044,656	B2 *	6/2015	Zhang	B62B 3/02
	9,254,858	B2	2/2016	Shellenberger	
	9,592,846	B1 *	3/2017	Hanson	B62B 7/068
	9,707,987	B2	7/2017	Hanson et al.	
	11,254,342	B2	2/2022	Kalinowski et al.	
	11,358,624	B2 *	6/2022	Guo	B62B 9/12
	11,479,285	B2	10/2022	Guo et al.	
	11,987,281	B2 *	5/2024	Guo	B62B 9/08
	2003/0204933	A1 *	11/2003	Yeh	B62B 7/04
					16/35 R
	2004/0226134	A1 *	11/2004	Chen	B60B 33/025
					16/35 R
	2005/0194755	A1 *	9/2005	Lan	B62B 7/042
					280/47.38
	2006/0071452	A1	4/2006	Yeh	
	2007/0013156	A1	1/2007	Wu et al.	
	2007/0085304	A1 *	4/2007	Yeh	B62B 9/085
					280/642
	2007/0262565	A1	11/2007	Bearup et al.	
	2010/0109271	A1 *	5/2010	Funakura	B62B 9/20
					280/47.36
	2010/0127481	A1 *	5/2010	Funakura	B62B 9/20
					280/658
	2010/0259022	A1 *	10/2010	Zhong	B60B 33/0042
					280/47.34
	2011/0285110	A1 *	11/2011	Xu-Hui	B62B 9/203
					280/650
	2012/0043730	A1 *	2/2012	Walther	B62K 5/08
					280/62
	2012/0056393	A1 *	3/2012	Funakura	B62B 7/04
					280/47.38
	2012/0326418	A1 *	12/2012	Ohnishi	B62B 7/066
					280/658
	2013/0069329	A1 *	3/2013	Chaudeurge	B60B 33/0073
					280/86
	2017/0057284	A1 *	3/2017	Wang	B62B 3/001
	2017/0066465	A1 *	3/2017	Hanson	B62B 9/082
	2017/0210407	A1	7/2017	Warwick et al.	
	2018/0065655	A1 *	3/2018	Leys	B62B 7/064
	2018/0162434	A1	6/2018	Kalinowski et al.	
	2020/0391783	A1 *	12/2020	Zheng	B62B 9/20
	2021/0046968	A1 *	2/2021	Zheng	B62B 9/087
	2021/0070344	A1 *	3/2021	Guo	B62B 9/12

CN	2737635	Y	11/2005
CN	2758149	Y	2/2006
CN	2827815	Y	10/2006
CN	201012618	Y	1/2008
CN	101678852	A	3/2010
CN	101687518	A	3/2010
CN	101973299	A	2/2011
CN	201761521	U	3/2011
CN	201923194	U	8/2011
CN	102248961	A	11/2011
CN	102673625	A	9/2012
CN	102923174	A	2/2013
CN	202753995	U	2/2013
CN	203111278	U	8/2013
CN	103568712	A	2/2014
CN	203681630	U	7/2014
CN	104044623	A	9/2014
CN	104118462	A	10/2014
CN	203888857	U	10/2014
CN	203946141	U	11/2014
CN	104176105	A	12/2014
CN	104781132	A	7/2015
CN	204871148	U	12/2015
CN	204956578	U	1/2016
CN	205632637	U	10/2016
CN	106143581	A	11/2016
CN	106608276	A	5/2017
CN	206351703	U	7/2017
CN	206885123	U	1/2018
CN	207293586	U	5/2018
CN	207790799	U	8/2018
CN	108820029	A	11/2018
CN	109305211	A	2/2019
CN	208682892	U	4/2019
CN	208915235	U	5/2019
CN	209617237	U	11/2019
EP	2719605	A1	4/2014
EP	2805868	A2	11/2014
EP	2946985	A1	11/2015
EP	2946985	B1	4/2018
JP	08067255	A	3/1996
JP	H09109604	A	4/1997
JP	20036054209	A	2/2003
JP	2008254693	A	10/2008
JP	2012136222	A	7/2012
JP	2014181031	A	9/2014
JP	2014213856	A	11/2014
JP	2015013600	A	1/2015
JP	3196047	U	2/2015
JP	2015091696	A	5/2015
JP	2015145144	A	8/2015
JP	2015209182	A	11/2015
JP	2017081252	A	5/2017
JP	3218610	U	10/2018
JP	2019077268	A	5/2019
JP	6651095	B1	2/2020
JP	7233398	B2	3/2023
TW	589261	B	6/2004
TW	M245120	U	10/2004
TW	201043510	A1	12/2010
TW	I434783	B	4/2014
TW	201434698	A	9/2014
TW	201710135	A	3/2017
TW	201711890	A	4/2017
TW	201914878	A	4/2019
TW	M606710	U	1/2021
WO	2013149642	A1	10/2013

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	2014110963 A1	7/2014
WO	2018196226 A1	11/2018

OTHER PUBLICATIONS

“Office Action for United States U.S. Appl. No. 17/743,338”, Mailed Date: Apr. 27, 2023, 23 pages.

“Office Action for United States U.S. Appl. No. 16/899,581”, Mailed Date: Aug. 2, 2022, 22 pages.

“First Office Action Issued in Corresponding Taiwanese Patent Application No. 112100025”, Mailed Date: Mar. 3, 2023, 12 pages.

“First Office Action Issued in Corresponding Taiwanese Patent Application No. 111137931”, Mailed Date: Jan. 30, 2023, 6 pages.

“First Office Action Issued in Corresponding Taiwanese Patent Application No. 109141101”, Mailed Date: Jul. 5, 2021, 12 pages.

“First Office Action Issued in Corresponding Taiwanese Patent Application No. 109131083”, Mailed Date: Oct. 13, 2021, 30 pages.

“First Office Action Issued in Corresponding Taiwanese Patent Application No. 109125001”, Mailed Date: Mar. 5, 2021, 5 pages.

“Search Report Issued in Corresponding Japanese Patent Application No. 2023-037646”, Mailed Date: Feb. 22, 2024, 37 pages.

“Notice of Reasons for Refusal Issued in Corresponding Japanese Patent Application No. 2023-037646”, Mailed Date: Feb. 27, 2024, 12 pages.

“Search Report Issued in Corresponding Japanese Patent Application No. 2023-024357”, Mailed Date: Oct. 31, 2023, 35 pages.

“First Office Action Issued in Corresponding Japanese Patent Application No. 2020-102214”, Mailed Date: Aug. 30, 2021, 6 pages.

“Second Office Action Issued in Corresponding Japanese Patent Application No. 2020-102214”, Mailed Date: Jun. 13, 2022, 8 pages.

“First Office Action Issued in Corresponding European Patent Application No. 20195526.7”, Mailed Date: Oct. 26, 2022, 4 pages.

“Extended European Search Report Issued in Corresponding European Patent Application No. 20179828.7”, Mailed Date: Nov. 11, 2020, 4 pages.

“Second Office Action Issued in Corresponding European Patent Application No. 20179828.7”, Mailed Date: Oct. 5, 2022, 4 pages.

“First Office Action Issued in Corresponding Chinese Patent Application No. 202080056990.4”, Mailed Date: Mar. 28, 2024, 22 pages.

“First Office Action Issued in Corresponding Chinese Patent Application No. 202010951546.0”, Mailed Date: Mar. 8, 2024, 17 pages.

“First Office Action Issued in Corresponding Chinese Patent Application No. 2020109500126”, Mailed Date: Aug. 2, 2022, 17 pages.

“Third Office Action Issued in Corresponding Chinese Patent Application No. 202010950012.6”, Mailed Date: Sep. 27, 2023, 13 pages.

“First Office Action Issued in Corresponding Chinese Patent Application No. 201911336477.6”, Mailed Date: Jan. 30, 2024, 14 pages.

“First Office Action Issued in Corresponding Chinese Patent Application No. 201910787608.6”, Mailed Date: May 8, 2021, 15 pages.

“Second Office Action Issued in Corresponding Chinese Patent Application No. 201910787608.6”, Mailed Date: Jan. 29, 2022, 15 pages.

“Third Office Action Issued in Corresponding Chinese Patent Application No. 201910787608.6”, Mailed Date: Apr. 18, 2022, 12 pages.

“First Office Action Issued in Corresponding Canadian Patent Application No. 3,151,487”, Mailed Date: Mar. 28, 2023, 5 pages.

“First Office Action Issued in Corresponding Australian Patent Application No. 2020293582”, Mailed Date: Mar. 6, 2023, 4 pages.

“Second Office Action Issued in Corresponding Australian Patent Application No. 2020293582”, Mailed Date: Nov. 6, 2023, 4 pages.

Office Action in corresponding Taiwanese patent application No. 112104479 issued May 3, 2023, and English translation of search report.

Notice of Allowance for corresponding Chinese Application No. 2019113364776 dated Dec. 13, 2024.

Notice of Allowance for corresponding U.S. Appl. No. 18/635,698 dated Jan. 7, 2025.

Notice of Allowance for corresponding Chinese Application No. 202080056990.4 dated Aug. 11, 2025, 4 pages.

First Examination Report for corresponding Australian Application No. 202385807 dated May 13, 2025, 4 pages.

* cited by examiner

100

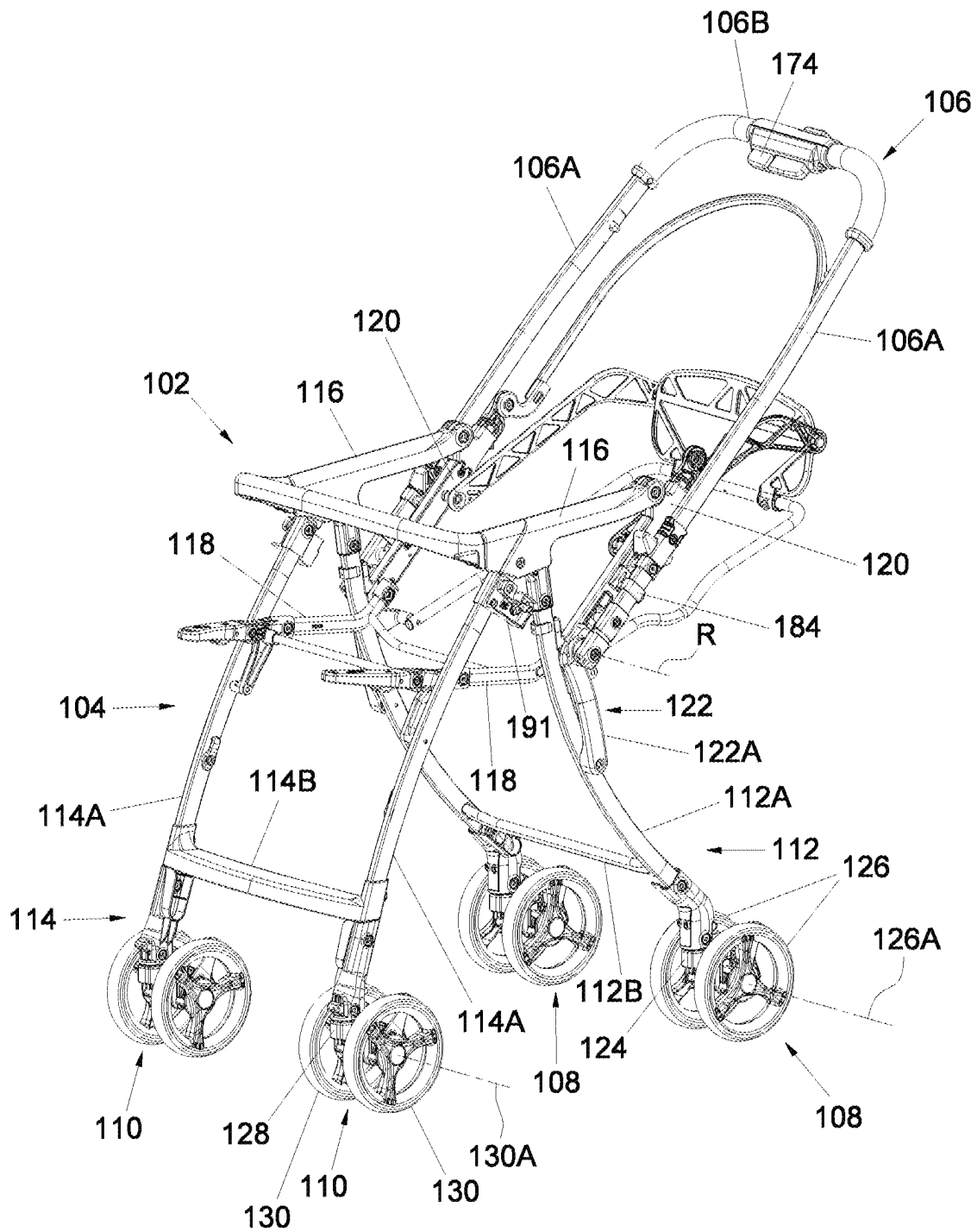


FIG. 1

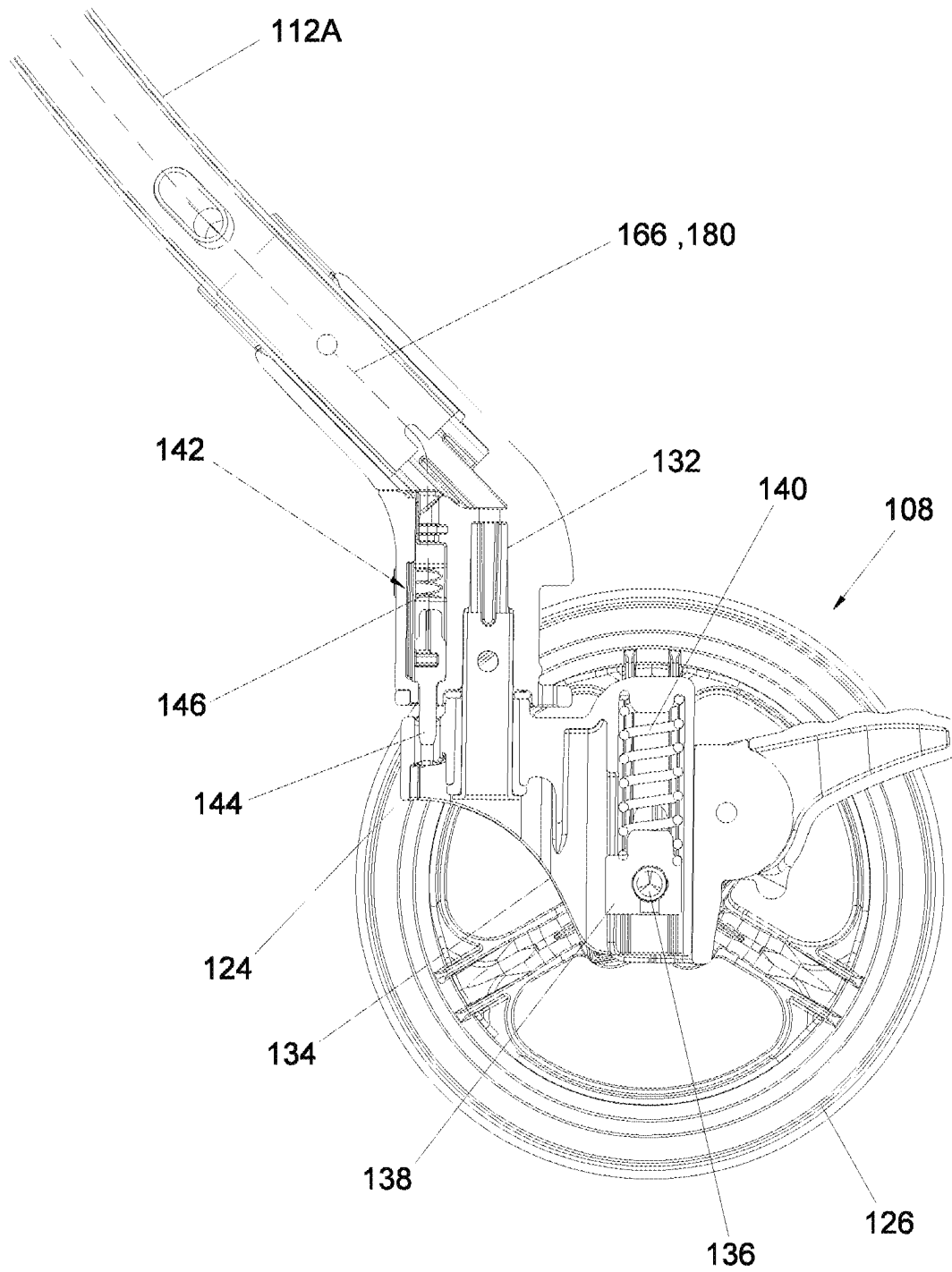


FIG. 2

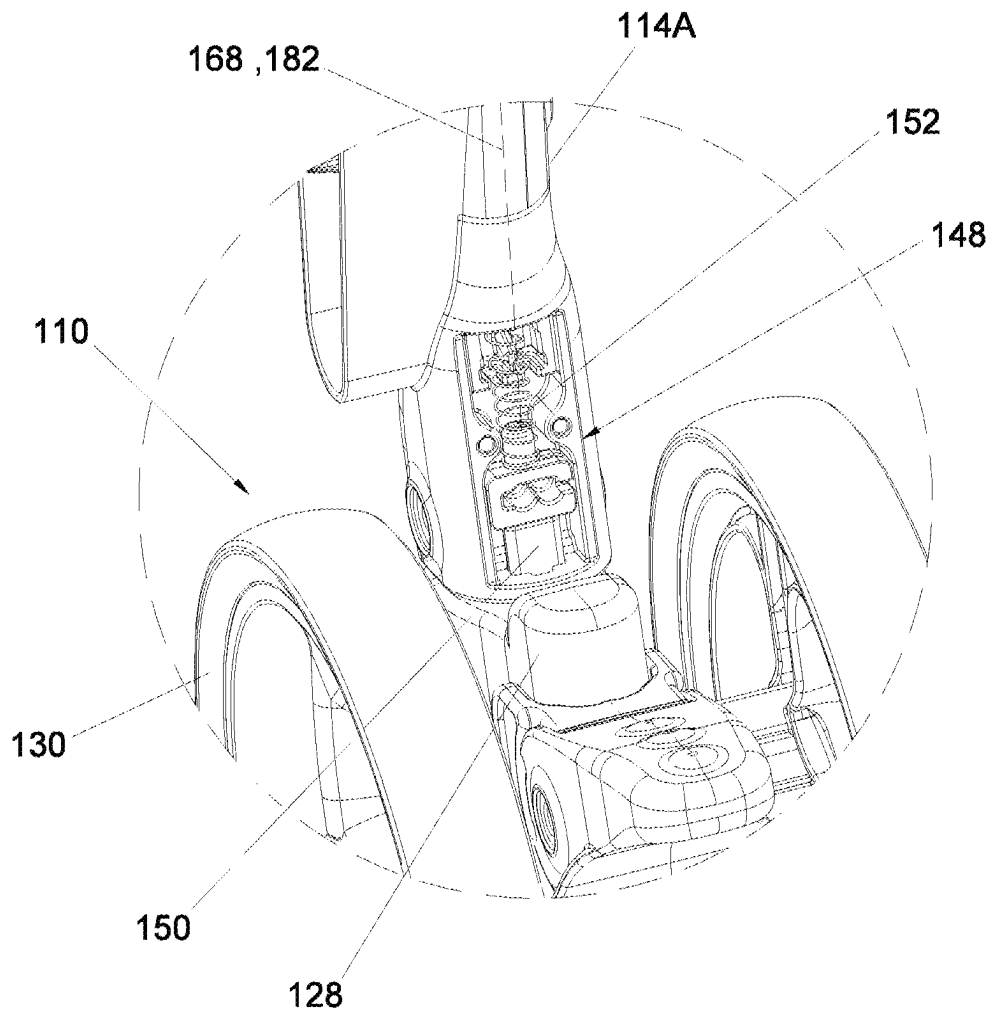


FIG. 3

100

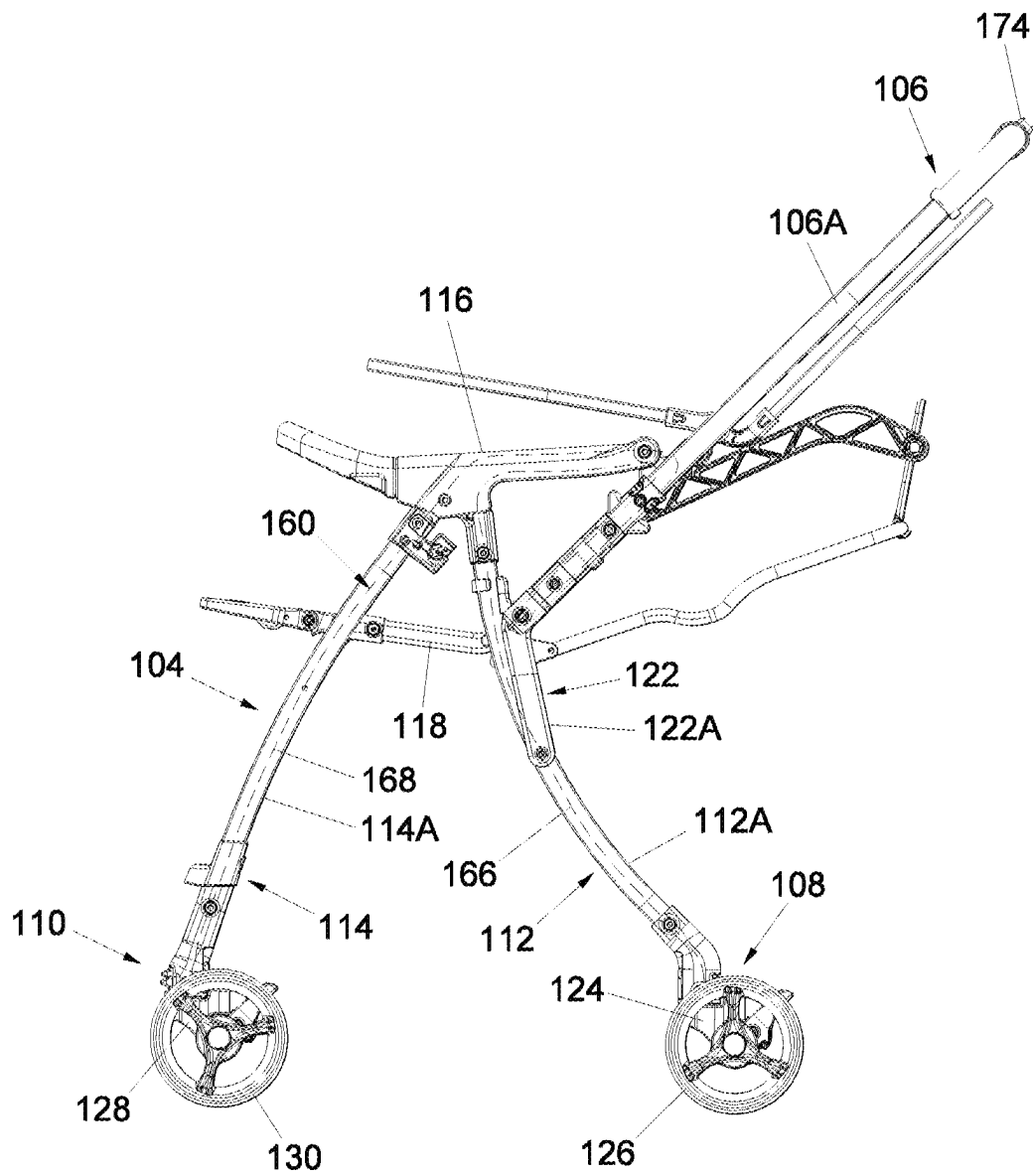


FIG. 4

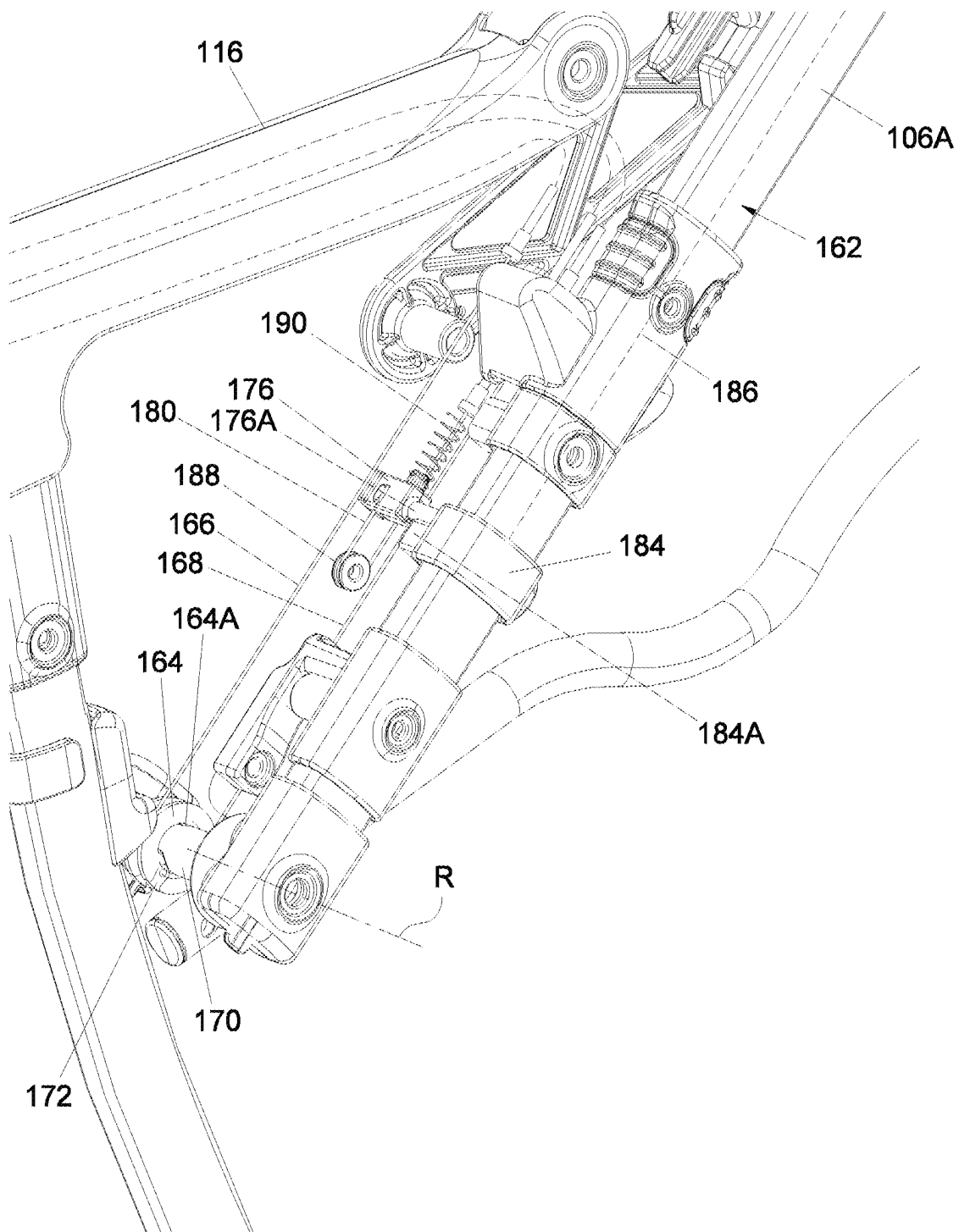


FIG. 5

100

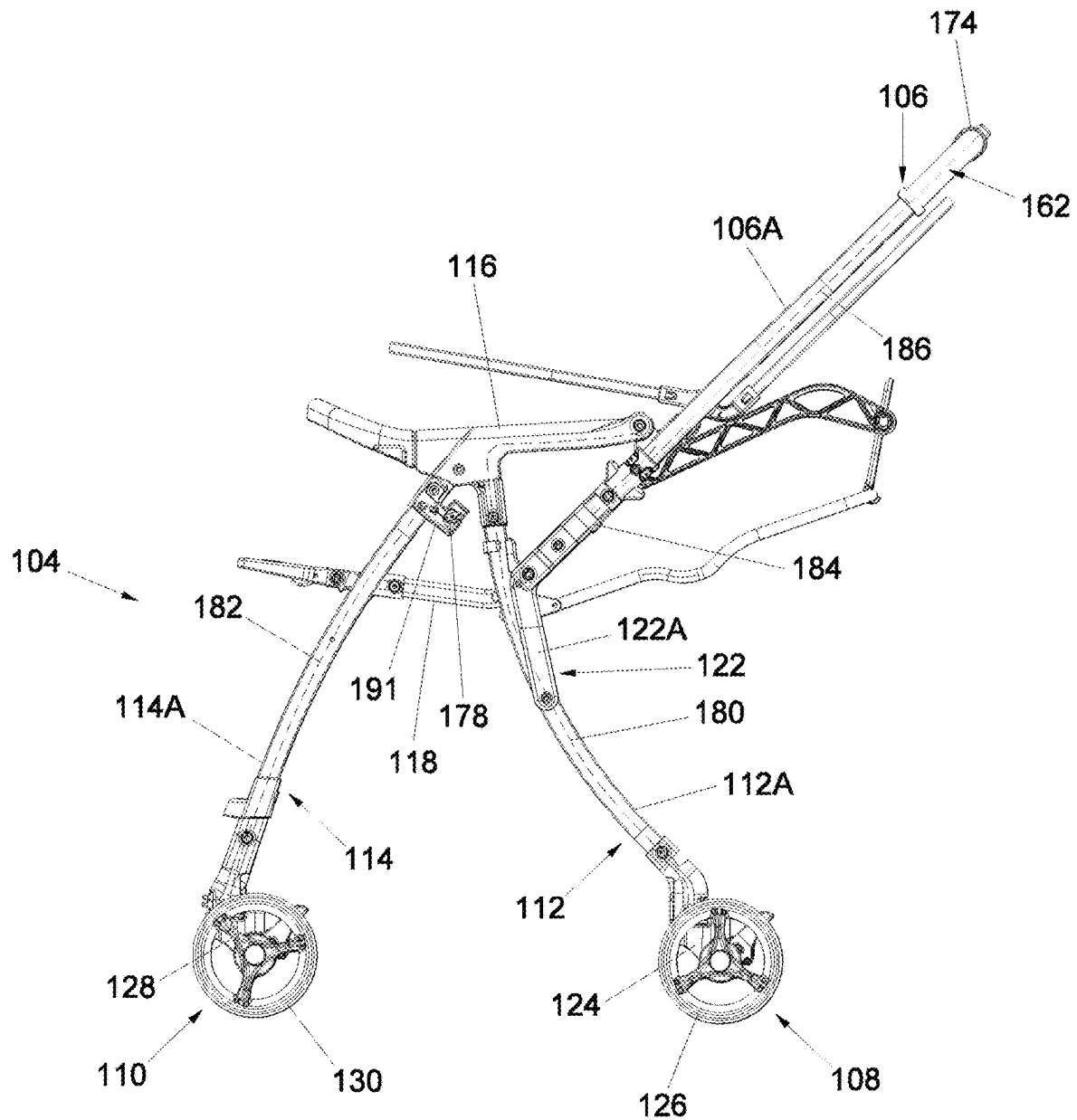


FIG. 6

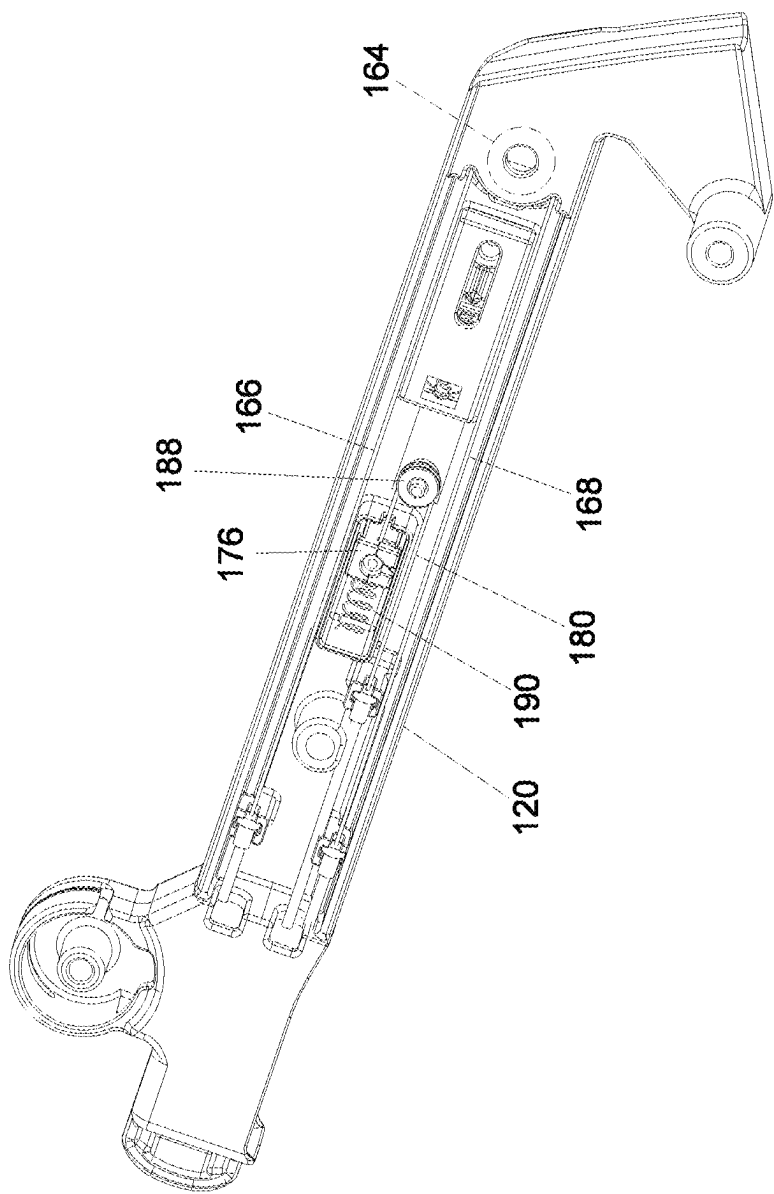


FIG. 7

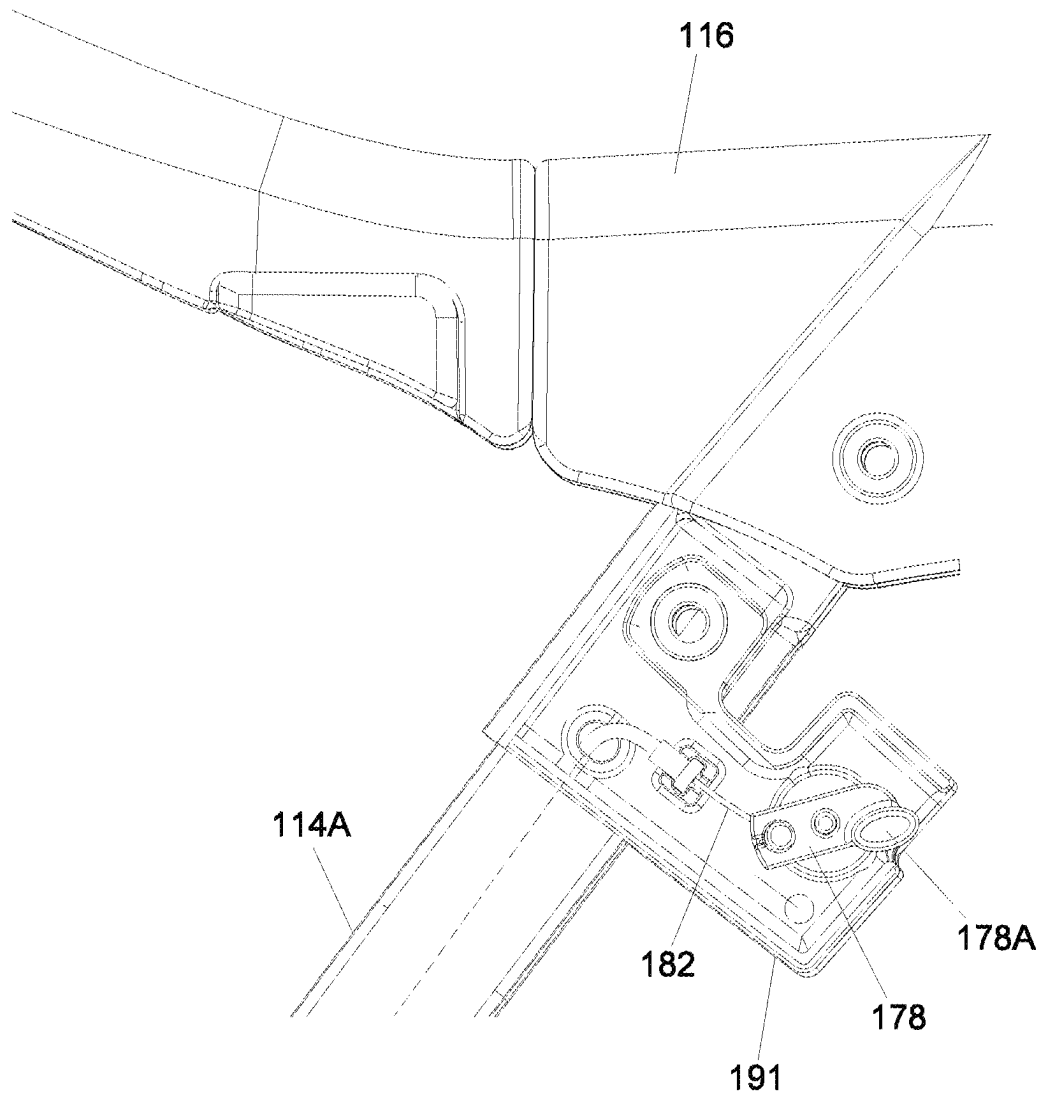


FIG. 8

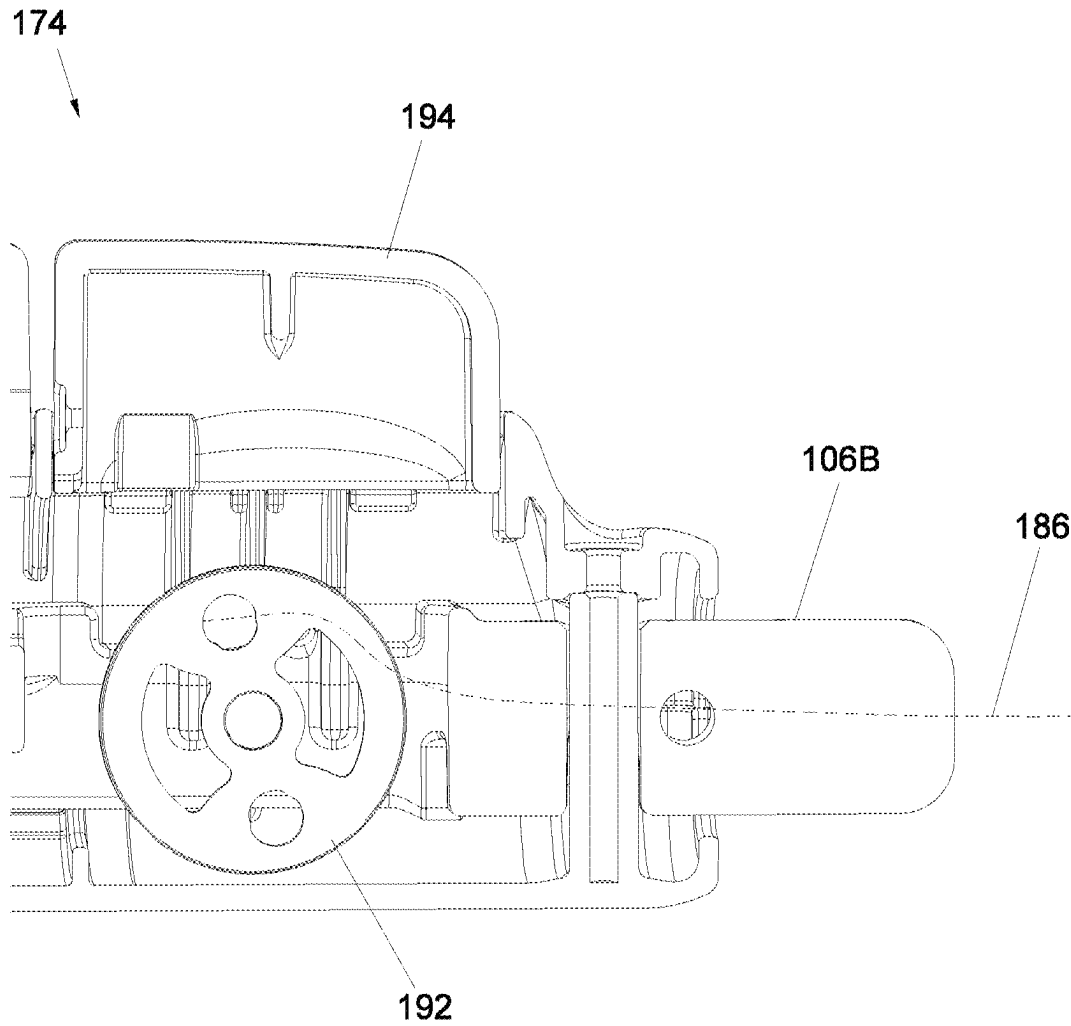


FIG. 9

100

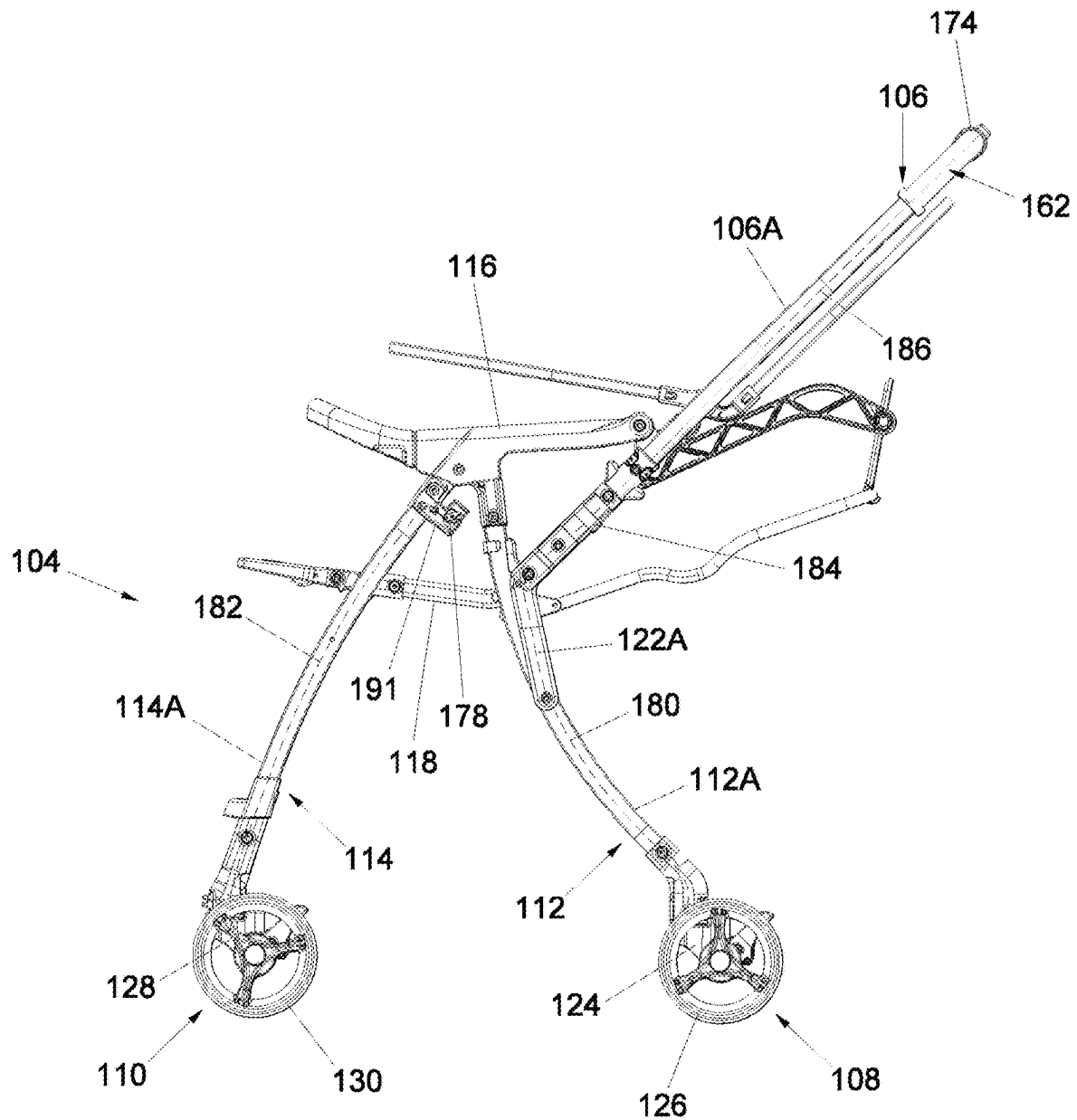


FIG. 10

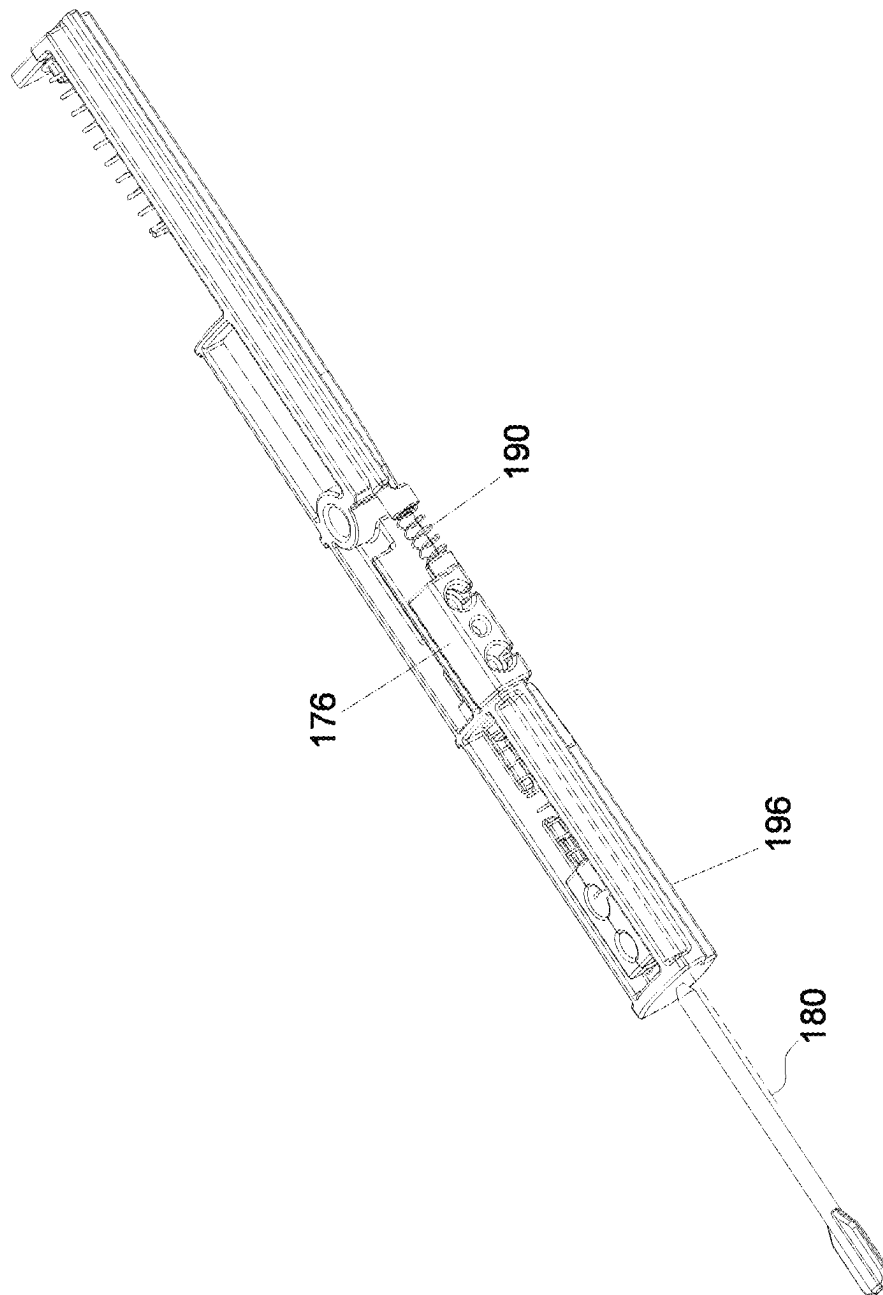


FIG. 11

1

WHEELED CARRYING APPARATUS**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a continuation application of U.S. patent application Ser. No. 17/945,323, filed on Sep. 15, 2022, which is a continuation application of U.S. patent application Ser. No. 17/017,401, filed on Sep. 10, 2020, now U.S. Pat. No. 11,479,285, issued on Oct. 25, 2022, which claims priority to Chinese Patent Application No. 201910858955.3, filed on Sep. 11, 2019, and to Chinese Patent Application No. 202010370191.6, filed on Apr. 30, 2020, which are incorporated by reference as if fully set forth.

FIELD OF THE INVENTION

The present invention relates to wheeled carrying apparatuses.

BACKGROUND OF THE INVENTION

Caregivers usually rely on a stroller apparatus to transport babies and children. Most stroller apparatuses typically have front wheels that can freely rotate horizontally relative to the stroller frame for changing the orientation of the wheel axes, and rear wheels that cannot rotate horizontally and have a fixed wheel axis oriented transversally. As a result, it may not be easy for the stroller apparatus to turn around in an environment of limited space.

Therefore, there is a need for an improved design that can offer better maneuverability and address at least the foregoing issues.

SUMMARY OF THE INVENTION

The present application describes a wheeled carrying apparatus having wheel assemblies that can be controllably locked and unlocked in horizontal orientation and can offer better maneuverability.

According to one embodiment, the wheeled carrying apparatus includes a standing frame, a handle frame, a first wheel mount, a locking assembly and a release mechanism. The standing frame and the handle frame are pivotally coupled to each other, the handle frame being rotatable relative to the standing frame between a first position and a second position. The first wheel mount carries a first wheel and is pivotally connected with the standing frame, the first wheel being rotatable relative to the first wheel mount about a first wheel axis. The locking assembly is connected with the standing frame, the locking assembly being configured to rotationally lock the first wheel mount with respect to the standing frame, and configured to unlock the first wheel mount so that the first wheel mount is rotatable relative to the standing frame for changing an orientation of the first wheel axis. The release mechanism includes a first portion provided on the standing frame and coupled to the locking assembly, and a second portion carried with the handle frame, the second portion being movable along with the handle frame during rotation of the handle frame between the first position and the second position. The second portion is operable to actuate the first portion and cause the locking assembly to unlock the locking assembly when the handle frame is in the first position, and the second portion is decoupled from the first portion so that the second portion is

2

inoperable to cause the locking assembly to unlock the first wheel mount when the handle frame is in the second position.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view illustrating an embodiment of a wheeled carrying apparatus;

FIG. 2 is a cross-sectional view illustrating one wheel assembly provided in the wheeled carrying apparatus;

FIG. 3 is an enlarged perspective view illustrating another wheel assembly provided in the wheeled carrying apparatus;

FIG. 4 is a side view illustrating some construction details of an actuating assembly provided in the wheeled carrying apparatus;

FIG. 5 is a perspective view illustrating some construction details of the actuating assembly and a release mechanism provided in the wheeled carrying apparatus;

FIG. 6 is a side view illustrating some construction details of the release mechanism provided in the wheeled carrying apparatus;

FIG. 7 is a perspective view illustrating a portion of the actuating assembly and release mechanism connected with a carrier bar in the frame structure of the wheeled carrying apparatus;

FIG. 8 is an enlarged view illustrating another portion of the release mechanism provided on a leg frame of the wheeled carrying apparatus;

FIG. 9 is an enlarged view illustrating construction details of an operating part provided on a handle frame of the wheeled carrying apparatus;

FIG. 10 is a side view illustrating a variant construction of the release mechanism;

FIG. 11 is a perspective view illustrating some construction details of the release mechanism shown in FIG. 10; and

FIG. 12 is a perspective view illustrating the wheeled carrying apparatus with the handle frame inclined in a second position different from that shown in FIG. 1.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The present application provides a wheeled carrying apparatus that includes a standing frame, and a plurality of wheel assemblies provided at a bottom of the standing frame. The wheel assemblies can be controllably locked and unlocked in horizontal orientation with respect to a standing frame, wherein an unlocked wheel assembly can rotate relative to the standing frame for changing the horizontal orientation of a wheel axis. According to the needs, some of the wheel assemblies are locked while others are unlocked, or all of the wheel assemblies can be unlocked at the same time as desired. Accordingly, the wheeled carrying apparatuses can be more flexible in use and have better maneuverability. Embodiments of the wheeled carrying apparatus described herein include child stroller apparatuses. However, it will be appreciated that wheeled carrying apparatuses that may incorporate the structures and features described herein can include, without limitation, child stroller apparatuses, trolleys, shopping carts, and the like. Accordingly, it is understood that the structures and features described herein may be generally applicable to any types of products having wheels, and the wheeled carrying apparatuses as described and claimed herein are not limited to the examples illustrated hereinafter.

FIG. 1 is a perspective view illustrating an embodiment of a wheeled carrying apparatus 100, which is exemplarily a

3

child stroller apparatus. Referring to FIG. 1, the wheeled carrying apparatus 100 can include a frame structure 102 comprised of a standing frame 104 and a handle frame 106 coupled to each other, and a plurality of wheel assemblies 108 and 110 provided at a bottom of the standing frame 104.

According to an example of construction, the standing frame 104 can include two leg frames 112 and 114, two side linking bars 116, two side support bars 118 and two carrier bars 120. The leg frame 112 can be exemplarily a rear leg frame, and the leg frame 114 can be exemplarily a front leg frame. Each of the leg frames 112 and 114 can respectively include two side segments disposed symmetrically at a left and a right side of the standing frame 104, e.g., two side segments 112A for the leg frame 112 and two side segments 114A for the leg frame 114, and a transversal segment fixedly connected with the two side segments, e.g., a transversal segment 112B fixedly connected with the two side segments 112A for the leg frame 112, and a transversal segment 114B fixedly connected with the two side segments 114A for the leg frame 114. At each of the left and right sides, the side segment 112A of the leg frame 112 and the side segment 114A of the leg frame 114 can be respectively coupled pivotally to one side linking bar 116, and one side support bar 118 can extend between side segments 112A and 114A and can be pivotally coupled to the side segment 114A, whereby the leg frames 112 and 114, the side linking bars 116 and the side support bars 118 can rotate relative to one another during folding and unfolding of the wheeled carrying apparatus 100.

The handle frame 106 can be respectively coupled pivotally to the standing frame 104 at the left and right sides via two bar linkages 122, wherein each bar linkage 122 can include a bar 122A. More specifically, the handle frame 106 can include two side portions 106A disposed symmetrically at the left and right sides, and a grip portion 106B connected with the two side portions 106A. At each of the left and right sides, the carrier bar 120 can have an upper portion pivotally connected with the side linking bar 116, and a lower portion respectively connected pivotally with the side support bar 118, the bar 122A of the bar linkage 122 and the side portion 106A of the handle frame 106 about a common pivot axis R. The bar 122A of the bar linkage 122 can be respectively connected pivotally with the side segment 112A of the leg frame 112, and with the side portion 106A of the handle frame 106 about the pivot axis R. With this construction, the handle frame 106 can rotate about the pivot axis R relative to the standing frame 104 and the bar linkage 122 for adjustment between the two positions of different inclinations so that the wheeled carrying apparatus 100 can be pushed in different directions with a child facing forward or rearward. For example, the handle frame 106 can be inclined toward the side of the leg frame 112 in a first position as shown in FIG. 1, and inclined toward the side of the leg frame 114 in a second position as shown in FIG. 12.

Referring to FIG. 1, each of the wheel assemblies 108 and 110 can include a wheel mount that is pivotally connected with the standing frame 104 and carries at least a wheel rotatable relative to the wheel mount about a wheel axis extending generally horizontally, whereby the wheel mount is rotatable relative to the standing frame 104 for changing an orientation of the wheel axis. For example, two wheel assemblies 108 can be provided at the lower ends of the side segments 112A of the leg frame 112, and two wheel assemblies 110 can be respectively provided at the lower ends of the two side segments 114A of the leg frame 114. Each wheel assembly 108 can include a wheel mount 124 that is pivotally connected with the leg frame 112 and carries at

4

least a wheel 126 rotatable relative to the wheel mount 124 about a wheel axis 126A extending generally horizontally, whereby the wheel mount 124 is rotatable for changing a horizontal orientation of the wheel axis 126A. Likewise, each wheel assembly 110 can include a wheel mount 128 that is pivotally connected with the leg frame 114 and carries at least a wheel 130 rotatable relative to the wheel mount 128 about a wheel axis 130A extending generally horizontally, whereby the wheel mount 128 is rotatable for changing a horizontal orientation of the wheel axis 130A. The wheel mounts 124 are spaced apart from the wheel mounts 128 along a lengthwise axis of the wheeled carrying apparatus 100 extending from a rear to a front thereof.

In conjunction with FIG. 1, FIG. 2 is a cross-sectional view illustrating one wheel assembly 108. Referring to FIGS. 1 and 2, the two wheel assemblies 108 can have a same construction. In each wheel assembly 108, the wheel mount 124 can be pivotally connected with the leg frame 112 about a pivot axle 132 so that the wheel mount 124 and the wheel 126 carried therewith can rotate in unison about the pivot axle 132. According to an example of construction, the wheel mount 124 can include a housing 134 that is pivotally connected with the leg frame 112 via the pivot axle 132. A wheel axle 136 defining the wheel axis 126A can be disposed through the housing 134 for pivotally connecting the wheel 126 with the wheel mount 124. According to an example of construction, the wheel axle 136 can be disposed through a sliding part 138 assembled inside the housing 134 for vertical sliding movement, and a spring 140 can be respectively connected with the sliding part 138 and the housing 134. The spring 140 can cushion a vertical displacement of the wheel 126 relative to the wheel mount 124 for dissipating undesirable shock energy during use.

Referring to FIG. 2, a locking assembly 142 is provided to rotationally lock and unlock the wheel mount 124 with respect to the standing frame 104. The locking assembly 142 can include a wheel mount latch 144 movably connected with the standing frame 104, and a locking spring 146 connected with the wheel mount latch 144. According to an example of construction, the wheel mount latch 144 can be slidably connected with the side segment 112A of the leg frame 112. More specifically, the side segment 112A can have a cavity, and the wheel mount latch 144 can have an elongate shape slidably received in the cavity of the side segment 112A, whereby the wheel mount latch 144 can slide to protrude outside the cavity or retract inward. The locking spring 146 can be disposed inside the cavity of the side segment 112A, and can have two ends respectively connected with the side segment 112A and the wheel mount latch 144. The locking spring 146 can bias the wheel mount latch 144 to protrude outward for engagement with the wheel mount 124.

In the locking assembly 142, the wheel mount latch 144 can slide downward to engage with the wheel mount 124 and thereby rotationally lock the wheel mount 124 with respect to the standing frame 104, and slide upward to disengage from the wheel mount 124 to unlock the wheel mount 124 so that the wheel mount 124 can rotate relative to the standing frame 104. The locking spring 146 can apply a biasing force for urging the wheel mount latch 144 to engage with the wheel mount 124. The same locking assembly 142 described above may be provided for each of the two wheel assemblies 108.

In conjunction with FIG. 1, FIG. 3 is an enlarged perspective view illustrating one wheel assembly 110. Referring to FIGS. 1 and 3, the wheel assembly 110 can have the same construction described previously, which can include

5

the wheel mount 128 pivotally connected with the leg frame 114 so that the wheel mount 128 and the wheel 130 carried therewith can rotate in unison relative to the leg frame 114. Moreover, a locking assembly 148 can be provided to rotationally lock and unlock the wheel mount 128 with respect to the standing frame 104. The locking assembly 148 may have a construction similar to the locking assembly 142 of the wheel assembly 108. According to an example of construction, the locking assembly 148 can include a wheel mount latch 150 movably connected with the standing frame 104, and a locking spring 152 connected with the wheel mount latch 150. For example, the wheel mount latch 150 may be slidably connected with the side segment 114A of the leg frame 114, and the locking spring 152 can have two ends respectively connected with the side segment 114A and the wheel mount latch 150. The wheel mount latch 150 can thereby slide downward to engage with the wheel mount 128 and thereby rotationally lock the wheel mount 128 with respect to the standing frame 104, and can slide upward to disengage and unlock the wheel mount 128 so that the wheel mount 128 can freely rotate relative to the standing frame 104 for changing the horizontal orientation of the wheel axis 130A. The locking spring 152 can bias the wheel mount latch 150 to engage with the wheel mount 128.

In conjunction with FIGS. 1-3, FIGS. 4-11 are schematic views illustrating further construction details that can be provided in the wheeled carrying apparatus 100. Referring to FIGS. 1-9, the wheeled carrying apparatus 100 can further include an actuating assembly 160 and a release mechanism 162.

Referring to FIGS. 1-5, the actuating assembly 160 is respectively coupled to the handle frame 106 and the wheel mount latches 144 and 150, and is operable independent of the release mechanism 162 to switch the wheel mount latches 144 and 150 between a locking and an unlocking state in accordance with the position of the handle frame 106. According to an example of construction, the actuating assembly 160 can include a link actuator 164 coupled to the standing frame 104, a linking part 166 coupling the wheel mount latch 144 to the link actuator 164, and a linking part 168 coupling the wheel mount latch 150 to the link actuator 164, the wheel mount latches 144 and 150 being respectively associated with the wheel assemblies 108 and 110 located at a same side (i.e., a left or right side) of the wheeled carrying apparatus 100. The linking parts 166 and 168 can include one or more cable and are illustrated with phantom lines FIG. 4.

The link actuator 164 can move in a first direction to pull the linking part 168 and urge the wheel mount latch 150 to disengage from the wheel mount 128 and concurrently relax the linking part 166 so that the wheel mount latch 144 can engage with the wheel mount 124 under the biasing force of the locking spring 146, and in a second direction opposite to the first direction to pull the linking part 166 and urge the wheel mount latch 144 to disengage from the wheel mount 124 and concurrently relax the linking part 168 so that the wheel mount latch 150 can engage with the wheel mount 128 under the biasing force of the locking spring 152. The link actuator 164 is movably linked to the handle frame 106 so that the link actuator 164 is movable in the first direction or the second direction in response to a rotation of the handle frame 106 relative to the standing frame 104 between the first position and the second position of different inclinations. In this embodiment, the link actuator 164 can be pivotally connected with the standing frame 104 about the pivot axis R, and can rotate as a unitary part relative to the standing frame 104 in the first and second direction. Accord-

6

ing to an example of construction, the side portion 106A of the handle frame 106 can be fixedly connected with a shaft portion 170 extending along the pivot axis R, and the link actuator 164 can be disposed adjacent to the carrier bar 120 and can be rotationally coupled to the handle frame 106 by engaging the shaft portion 170 through a hole 164A provided in the link actuator 164, whereby the link actuator 164 and the handle frame 106 can rotate in unison about the pivot axis R relative to the standing frame 104. Examples of suitable structures for the link actuator 164 may include, without limitation, a rotary plate, a ring and the like.

Referring to FIGS. 1-5, the two linking parts 166 and 168 can include cable portions that are provided as separate cables or are defined from a single cable, and can be routed along the standing frame 104. Suitable cables may include any elongate and flexible structures that can be tensioned for transmitting a drawing force, which may include, without limitation, metallic cables, cords, ropes, wires, strings, bands, strips, and the like. According to an example of construction, the two linking parts 166 and 168 can respectively include two cable portions that respectively couple the wheel mount latches 144 and 150 to the link actuator 164. For example, the linking part 166 can include a cable portion that is anchored to the link actuator 164 via an attachment part 172 and has an end anchored to the wheel mount latch 144, and the linking part 168 can include a cable portion that is anchored to the link actuator 164 via the attachment part 172 and has an end anchored to the wheel mount latch 150. The linking part 166 can exemplarily extend from the wheel mount latch 144 along the side segment 112A of the leg frame 112, the side linking bar 116 and the carrier bar 120 to the link actuator 164. The linking part 168 can exemplarily extend from the wheel mount latch 150 along the side segment 114A of the leg frame 114, the side linking bar 116 and the carrier bar 120 to the link actuator 164. The linking parts 166 and 168 may extend substantially parallel to each other along the carrier bar 120.

According to an example of construction, the two linking parts 166 and 168 may be defined from a single cable that can thereby form an integral single linking part, which can be routed along the frame structure like previously described, and can have a first and a second end that are respectively anchored to the wheel mount latch 144 and the wheel mount latch 150 and an intermediate portion that loops around the link actuator 164 and is anchored to the link actuator 164 via the attachment part 172.

In the actuating assembly 160, the link actuator 164 can thus rotate along with the handle frame 106 in a first direction to pull the linking part 168 (in particular the end thereof that is anchored to the wheel mount latch 150) and thereby urge the wheel mount latch 150 to disengage and unlock the wheel mount 128 and concurrently relax the linking part 166 (in particular the end thereof that is anchored to the wheel mount latch 144) so that the wheel mount latch 144 is allowed to engage and lock the wheel mount 124. Moreover, the link actuator 164 can rotate along with the handle frame 106 in a second direction opposite to the first direction to pull the linking part 166 (in particular the end thereof that is anchored to the wheel mount latch 144) and thereby urge the wheel mount latch 144 to disengage and unlock the wheel mount 124 and concurrently relax the linking part 168 (in particular the end thereof that is anchored to the wheel mount latch 150) so that the wheel mount latch 150 is allowed to engage and lock the wheel mount 128.

Since the two wheels assemblies 108 and 110 are provided at each of the left and right side of the wheeled

7

carrying apparatus 100, the actuating assembly 160 can be symmetrically arranged at the left and right side.

Referring to FIGS. 1-3 and 5-8, the release mechanism 162 includes an operating part 174 carried with the handle frame 106, and is configured so that the operating part 174 can be operatively coupled to one of the wheel mount latches 144 and 150 and decoupled from the other one of the wheel mount latches 144 and 150 depending on whether the handle frame 106 is in the first position inclined toward the side of the leg frame 112 or the second position inclined toward the side of the leg frame 114. For example, the operating part 174 is operable to cause the wheel mount latch 144 to disengage from the wheel mount 124 and meanwhile is operatively decoupled from the wheel mount latch 150 when the handle frame 106 is in the first position. When the handle frame 106 is in the second position, the operating part 174 is operable to cause the wheel mount latch 150 to disengage from the wheel mount 128 and meanwhile is operatively decoupled from the wheel mount latch 144. To this end, an embodiment of the release mechanism 162 can include the operating part 174, two link actuators 176 and 178 operable independently of each other, a linking part 180 coupling the wheel mount latch 144 to the link actuator 176, a linking part 182 coupling the wheel mount latch 150 to the link actuator 178, a driving part 184 carried with the handle frame 106, and a linking part 186 coupling the driving part 184 to the operating part 174. The linking parts 180, 182 and 186 may include cables and are illustrated with phantom lines in FIG. 6. Suitable cables may include any elongate and flexible structures that can be tensioned for transmitting a drawing force, which may include, without limitation, metallic cables, cords, ropes, wires, strings, bands, strips, and the like.

Referring to FIGS. 1-3 and 5-7, the link actuator 176 is movably connected with the standing frame 104, and can respectively move in one direction to pull the linking part 180 and thereby urge the wheel mount latch 144 to disengage from the wheel mount 124 and in an opposite direction to relax the linking part 180 so that the wheel mount latch 144 can engage with the wheel mount 124. The link actuator 176 can be placed at a position that is adjacent to the side portion 106A of the handle frame 106 when the handle frame 106 is in the first position. According to an example of construction, the link actuator 176 can be movably connected and carried with the carrier bar 120 of the standing frame 104, and can have a protrusion 176A projecting sideways. For example, the link actuator 176 may be slidably connected with the carrier bar 120, and can slide upward to pull the linking part 180 and thereby urge the wheel mount latch 144 to disengage from the wheel mount 124 and slide downward to relax the linking part 180 so that the wheel mount latch 144 can engage with the wheel mount 124.

The linking part 180 can include a cable, which may have one end anchored to the link actuator 176 and another end anchored to the wheel mount latch 144. The linking part 180 may extend substantially parallel to the two linking parts 166 and 168 of the actuating assembly 160 along a portion of the carrier bar 120.

Referring to FIGS. 5-7, the linking part 180 may extend along a portion of the side segment 112A of the leg frame 112, a portion of the side linking bar 116, a portion of the carrier bar 120, and can loop around a pulley 188 that is pivotally connected with the carrier bar 120. The pulley 188 may be disposed in a region of the carrier bar 120 that is located between the link actuator 164 of the actuating assembly 160 and the link actuator 176.

8

FIGS. 10 and 11 illustrate a variant construction in which the linking part 180 can operate like previously described but is routed along a different path. Referring to FIGS. 10 and 11, the linking part 180 may extend along a portion of the side segment 112A of the leg frame 112, the bar 122A and a portion of the carrier bar 120 for connecting with the link actuator 176. In this variant construction, the pulley 188 can be omitted, and the link actuator 176 can be connected with a coupling rod 196 provided on the carrier bar 120. The coupling rod 196 can slide along with the link actuator 176 to pull the linking part 180 and urge the wheel mount latch 144 to disengage from the wheel mount 124.

Referring to FIGS. 5 and 7, the release mechanism 162 can further include a spring 190 connected with the link actuator 176. For example, the spring 190 may have one end connected with the link actuator 176, and another end connected with an anchor structure fixedly connected with the carrier bar 120. The spring 190 can bias the link actuator 176 in a direction that urges the link actuator 176 in contact with the driving part 184.

In the variant construction illustrated in FIG. 11, the spring 190 can operate in a same manner, and may be connected between the link actuator 176 and an anchor structure provided on the coupling rod 196.

Referring to FIGS. 6 and 8, the link actuator 178 is movably connected with the standing frame 104, and can respectively move in one direction to pull the linking part 182 and thereby urge the wheel mount latch 150 to disengage from the wheel mount 128 and in an opposite direction to relax the linking part 182 so that the wheel mount latch 150 can engage with the wheel mount 128. The link actuator 178 can be placed at a position that is adjacent to the side portion 106A of the handle frame 106 when the handle frame 106 is in the second position. According to an example of construction, the link actuator 178 can be movably connected and carried with the side segment 114A of the leg frame 114, and can have a protrusion 178A projecting sideways. For example, the link actuator 178 may be pivotally connected with a bracket 191 that is fixedly attached to the side segment 114A of the leg frame 114, and can rotate in one direction to pull the linking part 182 and thereby urge the wheel mount latch 150 to disengage from the wheel mount 128 and rotate in an opposite direction to relax the linking part 182 so that the wheel mount latch 150 can engage with the wheel mount 128. It will be appreciated that the aforementioned pivotal connection is only an example of movably connecting the link actuator 178 with the side segment 114A of the leg frame 114, and other movable connections of the link actuator 178 may be suitable for pulling or relaxing the linking part 182. For example, a variant construction may have the link actuator 178 slidably connected with the side segment 114A of the leg frame 114 so that the link actuator 178 can slide for pulling or relaxing the linking part 182. According to an example of construction, the linking part 182 can include a cable, which may have one end anchored to the link actuator 178 and another end anchored to the wheel mount latch 150. According to an example of construction, the linking part 182 may extend along a portion of the side segment 114A of the leg frame 114 from the wheel mount latch 150 to the link actuator 178.

Referring to FIGS. 5-8, the driving part 184 is movably connected and carried with the handle frame 106, and is coupled to the operating part 174 via the linking part 186. According to an example of construction, the driving part 184 can be slidably connected with the side portion 106A of the handle frame 106 in a region below the side linking bar 116, and can have a protrusion 184A projecting sideways.

Accordingly, the driving part **184** is movable along with the handle frame **106** when the handle frame **106** rotates relative to the standing frame **104** between the first and second position of different inclinations. When the handle frame **106** is in the first position inclined toward the side of the leg frame **112**, the driving part **184** can be located adjacent to the link actuator **176** and displaced away from the link actuator **178**. When the handle frame **106** is in the second position inclined toward the side of the leg frame **114**, the driving part **184** can be located adjacent to the link actuator **178** and displaced away from the link actuator **176**.

According to an example of construction, the linking part **186** can include a cable, which may have one end anchored to the driving part **184** and another end anchored to the operating part **174**. For example, the operating part **174** may be provided on the grip portion **106B** of the handle frame **106**, and the linking part **186** can extend along the side portion **106A** of the handle frame **106** and can have two opposite ends respectively anchored to the operating part **174** and the driving part **184**. Accordingly, the operating part **174** can be operable to cause the driving part **184** to move relative to the side portion **106A** of the handle frame **106** and selectively urge either of the link actuator **176** and **178** in movement for unlocking either of the wheel mount latch **144** and **150**. For example, the operating part **174** is operable to cause the wheel mount latch **144** to disengage from the wheel mount **124** via a contact between the driving part **184** and the link actuator **176** when the handle frame **106** is in the first position, and the operating part **174** is operable to cause the wheel mount latch **150** to disengage from the wheel mount **128** via a contact between the driving part **184** and the link actuator **178** when the handle frame **106** is in the second position.

FIG. 9 is an enlarged view illustrating construction details of the operating part **174**. Referring to FIG. 9, the operating part **174** can include a barrel **192** and a button **194**. The barrel **192** can be pivotally connected with the grip portion **106B** of the handle frame **106**, and the linking part **186** can have an end anchored to the barrel **192**. The button **194** can be slidably connected with the grip portion **106B** and pivotally connected with the barrel **192**. With this construction, the button **194** can be pressed to urge the barrel **192** to rotate for pulling the linking part **186** so that the driving part **184** is urged in movement relative to the handle frame **106**.

Since the two wheels assemblies **108** and **110** are provided at each of the left and right side of the wheeled carrying apparatus **100**, the release mechanism **162** can be symmetrically arranged at the left and right side and likewise coupled to the operating part **174**.

In the wheeled carrying apparatus **100**, the handle frame **106** can be inclined toward the side of the leg frame **112** in the first position and toward the side of the leg frame **114** in the second position. The link actuator **164** of the actuating assembly **160** can rotate along with the handle frame **106** in the first direction for pulling the linking part **168** and relaxing the linking part **166** when the handle frame **106** rotates from the second position to the first position, which can cause the wheel mount latch **150** to disengage and unlock the wheel mount **128** and cause the wheel mount latch **144** to engage and lock the wheel mount **124**. Conversely, the link actuator **164** can rotate along with the handle frame **106** in the second direction for pulling the linking part **166** and relaxing the linking part **168** when the handle frame **106** rotates from the first position to the second position, which can cause the wheel mount latch **144** to

disengage and unlock the wheel mount **124** and cause the wheel mount latch **150** to engage and lock the wheel mount **128**.

While the handle frame **106** is in the first position and the operating part **174** is released, the wheel mount latch **144** is engaged with the wheel mount **124** to lock the wheel mount **124** to the standing frame **104**, and the wheel mount latch **150** is kept disengaged from the wheel mount **128** by a distance so that the wheel mount **128** is unlocked for free rotation relative to the standing frame **104**. For unlocking the wheel mounts **124** and **128** while the handle frame **106** is in the first position, a caregiver can actuate the operating part **174** to pull the linking part **186**, which causes the driving part **184** to slide upward and urge the link actuator **176** in movement against the biasing force of the spring **190** via a contact between the protrusion **184A** of the driving part **184** and the protrusion **176A** of the link actuator **176**. As a result, the link actuator **176** can pull the linking part **180** and thereby urge the wheel mount latch **144** to disengage from the wheel mount **124**. Meanwhile, the wheel mount latch **150** is not affected by the actuation of the operating part **174** and can remain in position disengaged from the wheel mount **128** by the same distance. Accordingly, all of the wheel mounts **124** and **128** can be unlocked at the same time in the first position of the handle frame **106**.

While the handle frame **106** is in the second position and the operating part **174** is released, the wheel mount latch **150** is engaged with the wheel mount **128** to lock the wheel mount **128** to the standing frame **104**, and the wheel mount latch **144** is kept disengaged from the wheel mount **124** by a distance so that the wheel mount **124** is unlocked for free rotation relative to the standing frame **104**. For unlocking the wheel mounts **124** and **128** while the handle frame **106** is in the second position, a caregiver can actuate the operating part **174** to pull the linking part **186**, which causes the driving part **184** to slide upward and urge the link actuator **178** in movement via a contact between the protrusion **184A** of the driving part **184** and the protrusion **178A** of the link actuator **176**. As a result, the link actuator **178** can pull the linking part **182** and thereby urge the wheel mount latch **150** to disengage from the wheel mount **128**. Meanwhile, the wheel mount latch **144** is not affected by the actuation of the operating part **174** and can remain in position disengaged from the wheel mount **124** by the same distance. Accordingly, all of the wheel mounts **124** and **128** can be unlocked at the same time in the second position of the handle frame **106**.

Since the operating part **174** is operatively coupled to only one of the wheel mount latches **144** and **150** at a time at each of the left and right side, the actuation for having all of the wheel mounts **124** and **128** unlocked at the same time would require a driving force that is relatively smaller. Therefore, less effort would be required to actuate the operating part **174**, and the component parts of the release mechanism **162** can be subjected to less mechanical stress.

Advantages of the wheeled carrying apparatuses described herein include the ability to lock and unlock wheel assemblies with respect to a standing frame in a flexible manner. According to the needs, some of the wheel assemblies are locked while others are unlocked, or all of the wheel assemblies are unlocked at the same time as desired. Accordingly, the wheeled carrying apparatuses can be more flexible in use and offer better maneuverability.

Realization of the wheeled carrying apparatuses has been described in the context of particular embodiments. These embodiments are meant to be illustrative and not limiting. Many variations, modifications, additions, and improve-

11

ments are possible. These and other variations, modifications, additions, and improvements may fall within the scope of the inventions as defined in the claims that follow.

What is claimed is:

1. A wheeled carrying apparatus comprising:
 - a standing frame and a handle frame pivotally coupled to each other, the handle frame being rotatable relative to the standing frame;
 - a first wheel mount carrying a first wheel and pivotally connected with the standing frame, the first wheel being rotatable relative to the first wheel mount about a first wheel axis;
 - a locking assembly connected with the standing frame, the locking assembly being configured to rotationally lock the first wheel mount with respect to the standing frame; and
 - a release mechanism synchronously moved with the handle to couple or decouple with the locking assembly, wherein the release mechanism includes a first portion provided on the standing frame and coupled to the locking assembly, and a second portion carried with the handle frame, the second portion being movable along with the handle frame during rotation of the handle frame between a first position and a second position;
 wherein the second portion is operable to actuate the first portion and cause the locking assembly to unlock the first wheel mount when the handle frame is in the first position, and the second portion is decoupled from the first portion so that the second portion is inoperable to cause the locking assembly to unlock the first wheel mount when the handle frame is in the second position.
2. The wheeled carrying apparatus according to claim 1, wherein causing the locking assembly to unlock the first wheel mount allows the first wheel mount to rotate relative to the standing frame for changing an orientation of the first wheel axis.
3. The wheeled carrying apparatus according to claim 1, wherein the second portion is located adjacent to the first portion so that the second portion is operable to actuate the first portion and cause the locking assembly to unlock the first wheel mount when the handle frame is in the first position, and the second portion is displaced away and disconnected from the first portion so that the second portion is inoperable to cause the locking assembly to unlock the first wheel mount when the handle frame is in the second position.
4. The wheeled carrying apparatus according to claim 1, wherein the locking assembly includes a first wheel mount latch movably connected with the standing frame, the first wheel mount latch being movable to engage with the first wheel mount for rotationally locking the first wheel mount with respect to the standing frame and to disengage from the first wheel mount for rotation of the first wheel mount relative to the standing frame.
5. The wheeled carrying apparatus according to claim 1, wherein the release mechanism is symmetrically arranged on the handle frame.

12

6. The wheeled carrying apparatus according to claim 1, wherein the release mechanism is coupled to a distal end of a side portion of the handle frame.

7. The wheeled carrying apparatus according to claim 6, wherein the release mechanism extends in a direction parallel to an extension direction of the side portion of the handle frame.

8. A wheeled carrying apparatus comprising:

a standing frame and a handle frame pivotally coupled to each other, the handle frame being rotatable relative to the standing frame;

a first wheel mount carrying a first wheel and pivotally connected with the standing frame, the first wheel being rotatable relative to the first wheel mount about a first wheel axis;

a locking assembly connected with the standing frame, the locking assembly being configured to rotationally lock the first wheel mount with respect to the standing frame; and

a release mechanism synchronously moved with the handle to couple or decouple with the locking assembly, wherein the release mechanism includes a first portion provided on the standing frame and coupled to the locking assembly, and a second portion carried with the handle frame, the second portion being movable along with the handle frame during rotation of the handle frame between a first position and a second position;

the first portion includes a link actuator that is coupled to the locking assembly via a first linking part, and the second portion includes a driving part and an operating part that are coupled to each other via a second linking part, the driving part being located adjacent to the link actuator when the handle frame is in the first position and is displaced away from the link actuator when the handle frame is in the second position.

9. The wheeled carrying apparatus according to claim 8, wherein the driving part is movably connected with a side portion of the handle frame.

10. The wheeled carrying apparatus according to claim 8, wherein the handle frame has a side portion and a grip portion connected with each other, the driving part being movably connected with the side portion, and the operating part being provided on the grip portion.

11. The wheeled carrying apparatus according to claim 8, wherein any one of the first linking part and the second linking part includes a cable.

12. The wheeled carrying apparatus according to claim 8, wherein the first linking part loops around a pulley.

13. The wheeled carrying apparatus according to claim 8, wherein the link actuator is connected with a spring, the spring biasing the link actuator in a direction that urges the link actuator in contact with the driving part.

14. The wheeled carrying apparatus according to claim 8, wherein the standing frame includes a carrier bar pivotally connected with the handle frame, the link actuator being movably connected and carried with the carrier bar.

* * * * *